

Backlash Control in Robotic Manipulators

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1. Introduction

Industrial robotic arms depend on accurate joint control to achieve precise motion in tasks like assembly, welding, and pick-and-place. Most joints are driven by high-ratio gear transmissions, which give out large output torque but also introduce mechanical problems. One of which is *backlash*: the free rotational play which is caused by a gap between parts that make contact.

Backlash creates an angular "dead-zone" or gap between the motor shaft and the load shaft. When the displacement lies within this gap, the motor moves without transmitting torque to the load. Only when the gear tooth pair comes into contact does the torque re-engage. In practice, even 1° - 2° of backlash can cause lost motion during reversing direction, path deviation, and low-speed vibration.

From a controls perspective, backlash is challenging because it introduces a piecewise, nonsmooth, and state-dependent nonlinearity. Inside the dead-zone the input torque to the load is zero, whereas outside the dead-zone the motor and load are coupled through an elastic shaft. Conventional linear controllers can cause oscillations around the dead-zone and struggle to meet tight tracking requirements.

The goal of this project is to investigate control strategies for a single robot joint with backlash, modeled as a two-mass motor-load system. The joint is a SISO system with motor torque as the input and load angle as the output. Our group designs and compares several controllers: linear quadratic regulator (LQR), second-order pole placement, linear model predictive control (MPC), and nonlinear MPC (NMPC).

2. Modeling of the Robot Joint

2.1 Physical system overview

The plant is a motor-load two-mass system with backlash. The motor drives the load through a shaft that includes a dead-zone.

Motor side: rotor inertia $J_m = 0.02 \text{ kg}\cdot\text{m}^2$

Load side: link inertia $J_l = 0.05 \text{ kg}\cdot\text{m}^2$

Transmission: shaft stiffness $K_s = 5 \text{ N}\cdot\text{m}/\text{rad}$, shaft damping $B_s = 0.05 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$, and backlash half-width $d = 0.03 \text{ rad}$ ($\approx 1.7^\circ$).

2.2 SISO system definition

The joint is modeled as a SISO system:

Input (single): motor torque command $u(t)$ [$\text{N}\cdot\text{m}$], limited by actuator constraints $-5 \leq u(t) \leq 5$.

Output (single): load angular position $y(t) = \theta_l(t)$ [rad], measured as output.

Internally, the system has four states (two on the motor side and two on the load side), but only the load angle is fed back to the controller.

2.3 State variables

We used the following state vector:

$$\mathbf{x}(t) = [\theta_m(t), \dot{\theta}_m(t), \theta_l(t), \dot{\theta}_l(t)]^T$$

The relative displacement and velocity between motor and load,

$$\text{gap}(t) = \theta_m(t) - \theta_l(t), \quad \dot{\text{gap}}(t) = \dot{\theta}_m(t) - \dot{\theta}_l(t)$$

determine whether the transmission is engaged or in the backlash dead-zone.

2.4 Backlash and shaft torque model

The torque transmitted through the shaft, $\tau_s(t)$, is modeled as a dead-zone nonlinearity followed by a linear spring-damper:

$$\tau_s(t) = 0, \quad \text{if } |\theta_m - \theta_l| \leq d \quad (\text{dead-zone, disengaged})$$

$$\tau_s(t) = K_s(\theta_m - \theta_l - d \cdot \text{sign}(\theta_m - \theta_l)) + B_s(\dot{\theta}_m - \dot{\theta}_l), \quad \text{if } |\theta_m - \theta_l| > d \quad (\text{engaged})$$

2.5 State-Space Representation and Linearization

For controller design we linearize the system in the *engaged* region, where the shaft behaves like a linear spring-damper. In this region we approximate

$$\tau_s \approx K_s(\theta_m - \theta_l) + B_s(\dot{\theta}_m - \dot{\theta}_l)$$

leading to the following continuous-time linear model

$$\dot{x}(t) = \mathbf{A} x(t) + \mathbf{B} u(t), \quad y(t) = \mathbf{C} x(t)$$

where the system matrices $\mathbf{A}$ and $\mathbf{B}$ are given by:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_s}{J_m} & -\frac{B_s}{J_m} & \frac{K_s}{J_m} & \frac{B_s}{J_m} \\ 0 & 0 & 0 & 1 \\ \frac{K_s}{J_l} & \frac{B_s}{J_l} & -\frac{K_s}{J_l} & -\frac{B_s}{J_l} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{J_m} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 0 \quad 1 \quad 0], \quad \mathbf{D} = 0.$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -250 & -2.5 & 250 & 2.5 \\ 0 & 0 & 0 & 1 \\ 100 & 1 & -100 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 50 \\ 0 \\ 0 \end{bmatrix}$$

3. Analysis of System properties

3.1 Stability

The stability of our linear system is determined by the eigenvalues of the matrix A. Numerical computation through MATLAB gives the following eigenvalues in the figure as the solutions.

$$\lambda(A) = \{0, 0, -1.75 \pm j 18.6\}.$$

The system contains two eigenvalues at 0 and 2 eigenvalues with negative real components. If the algebraic and geometric multiplicities of the 0 eigenvalue are equal then the system is marginally stable, otherwise it is unstable. The algebraic multiplicity of an eigenvalue is how many times it appears as a root of the characteristic polynomial. From the set above, $\lambda=0$ appears twice, so its algebraic multiplicity is 2. The geometric multiplicity measures how many independent eigenvectors are associated with that eigenvalue. This can be found by determining the dimensions of the null-space of A. After calculations the geometric multiplicity is only 1, meaning that **the open loop system is unstable** where certain initial conditions lead to unbounded growth in the states over time. Together, these characteristics confirm that the linearized engaged model is open-loop unstable and unsuitable for tracking or disturbance rejection without feedback.

3.2 Controllability

The controllability matrix has full rank (rank = 4), so the single motor torque input can influence all states.

$$P = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix} = \begin{bmatrix} 0 & 50 & -150 & -11925 \\ 50 & -150 & -11925 & 97750 \\ 0 & 0 & 50 & 4790 \\ 0 & 50 & 4790 & -37673 \end{bmatrix} \Rightarrow \text{rank}(P) = 4$$

3.3 Observability

The observability matrix is also full rank (rank = 4), meaning the full state is, in principle, reconstructible from measurements of the load angle alone. These properties justify the use of state-space methods and MPC for this system.

$$Q = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 100 & 1 & -100 & -1.2 \\ -370 & 95.8 & 370 & -96.06 \end{bmatrix} \Rightarrow \text{rank}(Q) = 4$$

4. Simulation

4.1 Second Order Approximation

Since the 4th order system is fully controllable, all four closed-loop poles can be placed. If two poles are placed 3-5x farther away from the origin than the other two, the system can be approximated as a second order system with the two closer poles.

Oscillations will cause the system to repeatedly pass through the backlash deadzone, which will cause damage and may also cause the output load angle to continuously oscillate within the deadzone instead of reaching the steady state goal. To avoid oscillations, the overshoot must be zero, meaning the closed loop system must be critically damped or overdamped. Critically damped would be the fastest, but $\zeta=1.001$ was chosen to avoid placing with high multiplicity, which would cause errors in code. For a fast response, the desired natural frequency is increased until the maximum input reaches the torque limit of $|u| < 5$. $\omega_n = 7$. The first two poles will be the solutions to $0 = s^2 + 2\zeta\omega_n s + \omega_n^2$.

$$\lambda_{des,02} = [-6.694 \quad -7.320 \quad -22.180 \quad -22.400]$$

$$K_{O2} = [14.631 \quad 1.102 \quad -9.762 \quad 0.679] \quad G_{O2} = 4.869$$

4.2 Linear Quadratic Regulator

This control method is based on the relative costs of each state and the control input. Then, a controller is chosen which minimizes the integral of cost until the system reaches steady state. For this method, states 1 and 2, the motor angle and angular velocity, have little to no associated cost. However, the input is “cheap” relative to states 3 and 4, the load angle and angular velocity. Therefore, the optimal control strategy can be considered “cheap.” To increase the speed of the response, Q_{33} was increased until the maximum input reached the limit. Although this controller was not designed with overshoot in mind, it also gave zero overshoot.

$$Q = \begin{bmatrix} 0.01 & 0 & 0 & 0 \\ 0 & 0.01 & 0 & 0 \\ 0 & 0 & 25 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R = 1$$
$$K_{LQR} = [5.549 \quad 0.475 \quad -0.548 \quad 1.030] \quad G_{LQR} = 5.001$$

4.3 Observer Design

Since the system is observable, an observer can be used to estimate the actual states from a single sensor outputting the load angle. Since the controller will use the estimated states instead of the actual states, the observer must be faster than the controller, quickly converging the estimation error to zero. This can be designed by placing negative real observer poles 3-5 times farther from the origin than all controller poles. An observer was created for each of the above controllers, which did not change the reference gain G . Ideally, the observers should not change the response of the system much.

$$L_{O2} = [10140 \quad 157120 \quad 270 \quad 27160]^T \quad L_{LQR} = [737 \quad -3431 \quad 118 \quad 4759]^T$$

In the following simulations with an observer, the initial error is as follows. There is no error in state 3 because the sensor directly measures this state.

$$e_0 = [-0.2 \quad -0.2 \quad 0 \quad -0.2]^T$$

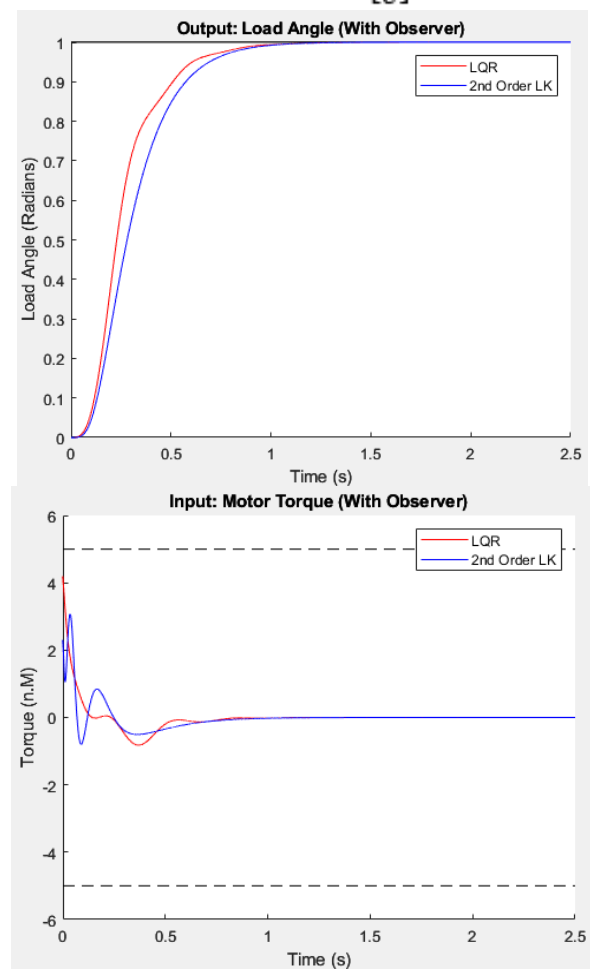
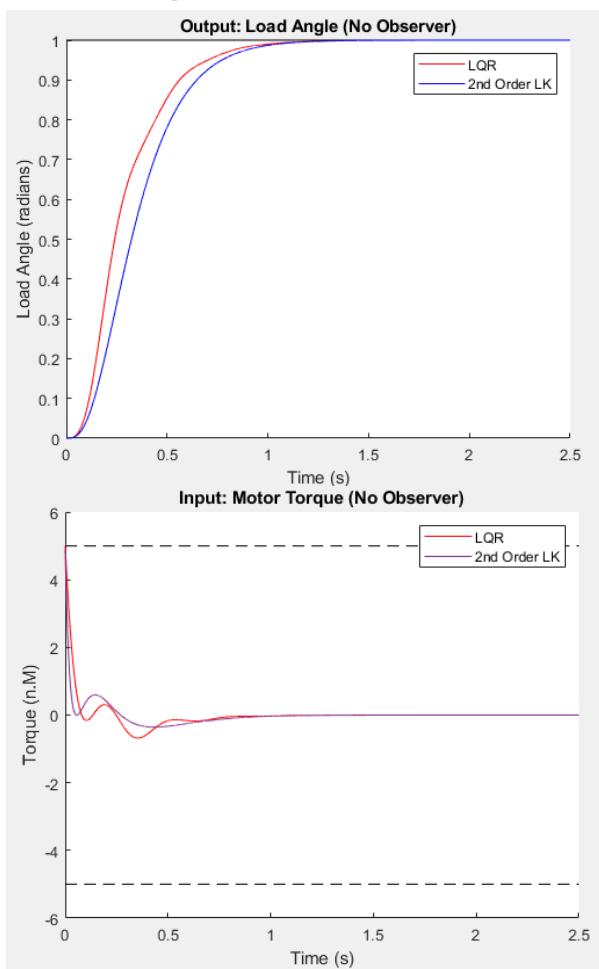
The closed loop systems were calculated in matlab using ss() and lsim() based on the following closed loop models.

$$\dot{x} = (A - BK)x + BGu$$

$$y = C - DK + DGu$$

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix} + \begin{bmatrix} BG \\ 0 \end{bmatrix} u$$

$$y = [C - DK \quad DK] \begin{bmatrix} x \\ e \end{bmatrix} + DGu$$



The above plots show the closed loop responses of the system with the controllers and observers designed above. Both controllers quickly reach steady state at the reference signal with no overshoot, and the input torque also stays within the limit. LQR is better than the second-order approximation, matching “cheap” control strategy. The responses with an

observer have very similar responses to no observer (could also be interpreted as an ideal observer). The controllers and observers meet all the design requirements.

[ControllerObserverCode](#)

4.4 Linear MPC Design

To evaluate predictive, constraint-aware controller performs on the modelled system, we designed a linear Model Predictive Controller (MPC) from the linearized engaged model from section 2.5. The goal was to regulate the load angle to desired trajectory while remaining within input limits and avoiding aggressive torque changes. The discrete model

$$x_{k+1} = A_d x_k + B_d u_k, y_k = C x_k$$

is used for prediction inside the MPC. The controller minimizes tracking error, control effort and input rate changes over a prediction horizon $N_p = 40$ and control horizon $N_c = 10$. The tuning weights were

$$Q = 50, R = 0.01, R_d = 0.1$$

The input torque was constrained to , torque limits $-1 \leq u_{k+j} \leq 1$ Nm. This model assumes the system stays engaged, so backlash is not included in its prediction model.

Performance(Linear Plant):

RMS load tracking error: $e_{\text{RMS}} \approx 0.0867$ rad,

Overshoot $\approx 7.54\%$

Performance(Nonlinear Backlash Plant):

RMS error increases to 0.1007 rad

Overshoot remains similar

System in the dead-zone $\approx 45\%$ of the time

The main limitation is that the linear MPC cannot anticipate the dead zone which leads to prediction mismatch, delayed torque transfer. Even with this, it still achieves smoother inputs and better tracking than basic linear controller.

[LinearMPC MATLAB code](#)

4.5 Nonlinear MPC Design

The nonlinear MPC (NMPC) has nonlinear dynamics inside the prediction model. At each sample k the NMPC solves over a prediction horizon $N_p = 50$ (0.5 s) and control horizon $N_c =$

8. The cost function penalizes tracking error, control effort, and control rate changes:

$$J = \sum (\mathbf{x}_{k+j|k} - \mathbf{r}_{k+j})^T \mathbf{Q} (\mathbf{x}_{k+j|k} - \mathbf{r}_{k+j}) + R u_{k+j}^2 + R_d (u_{k+j} - u_{k+j-1})^2$$

subject to the nonlinear dynamics and rate limits $|u_{k+j} - u_{k+j-1}| \leq 0.05$.

We test the NMPC on a step command in load angle from 0 to 0.5 rad over a simulation horizon of $T_{\text{sim}} = 6$ s ($N_{\text{sim}} = 600$ steps).

Key performance metrics are:

RMS load tracking error: $e_{\text{RMS}} \approx 0.102$ rad,

steady-state error: ≈ 0.008 rad,

settling time (within 2% band): ≈ 0.88 s,

overshoot: $\approx 3.5\%$,

fraction of time in the engaged region: $\approx 15\%$.

The position plot shows that the load quickly reaches the reference with little overshoot and then converges to within a small steady-state error. The control input is smooth, and is within the torque limit. The gap plot demonstrates the core benefit of the NMPC formulation: after the initial transient, the controller keeps the motor-load gap just outside the dead-zone

boundaries $\pm d$. The transmission is mostly engaged, which prevents the oscillations and repeated disengagements observed with purely linear MPC designs.

[NMPC MATLAB code](#)

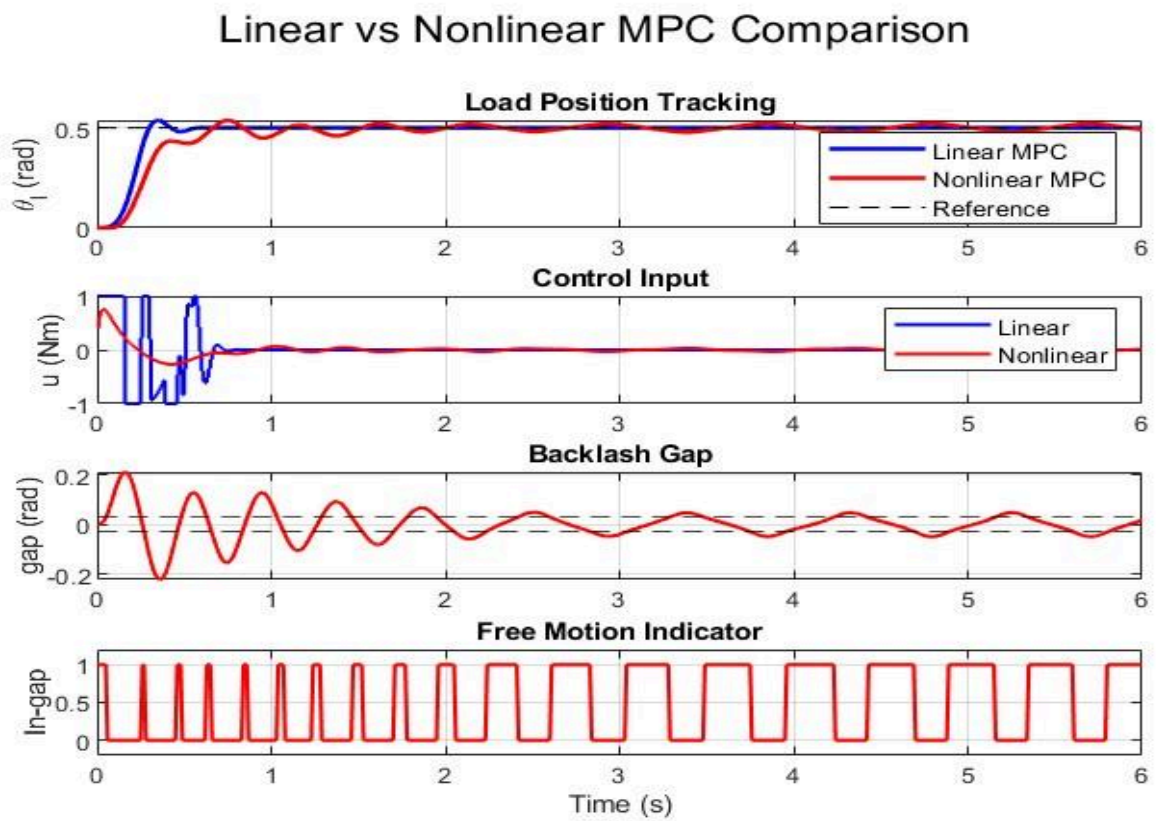


Figure: linear MPC simulation results

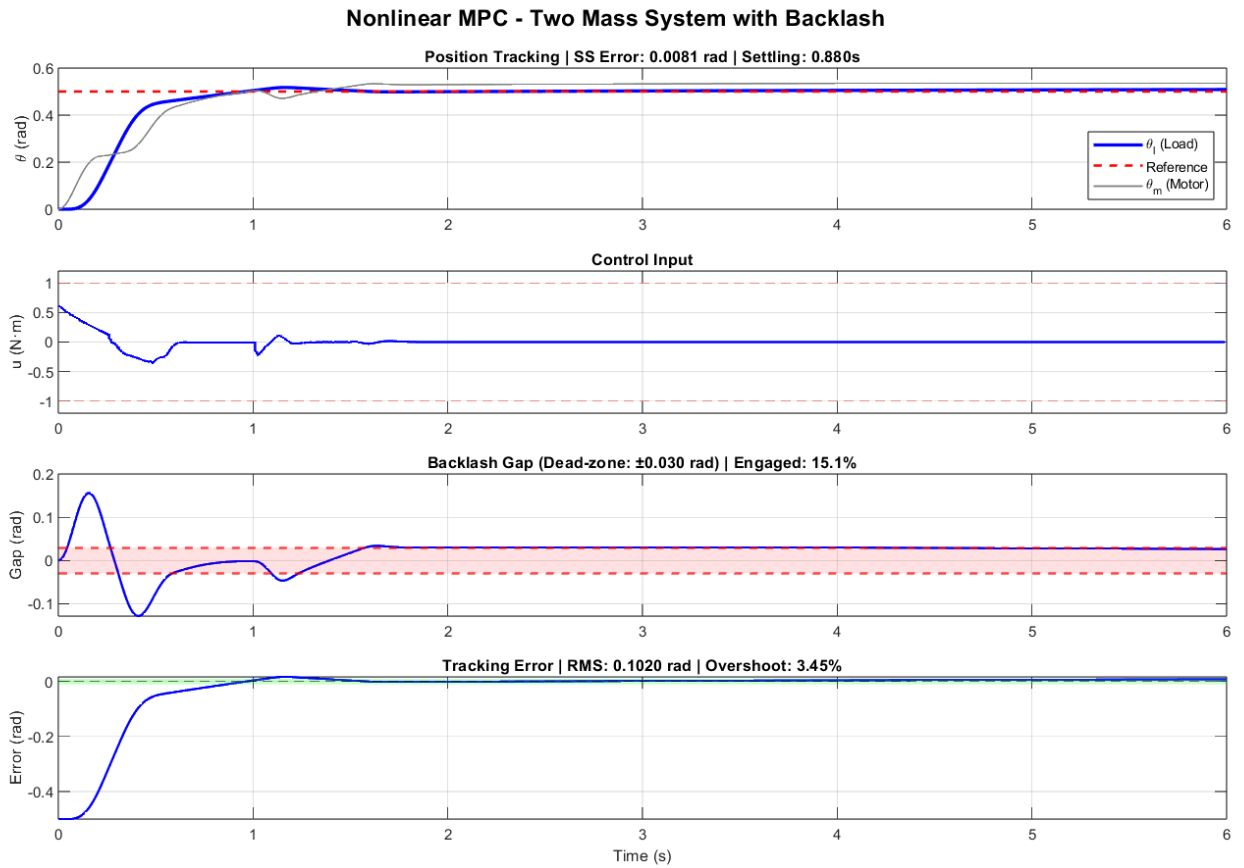


Figure: Nonlinear MPC simulation results

5. Conclusion and Discussion

Summary

In this project we compared four controllers for a robot joint with backlash: pole placement, LQR, linear MPC and nonlinear MPC. The linear controllers worked well on the linear model but had trouble when the backlash dead zone appeared. Linear controllers spent a lot of time inside the dead zone which lowered the tracking accuracy. The nonlinear MPC gave the best result because it uses the full nonlinear model, kept the system mostly engaged, and reached the goal with small overshoot and fast settling.

Major Design Decisions

A main choice was to linearize the model in the engaged region because that is where we want the joint to stay. If we linearized around the dead zone the controller would not try to avoid it. We also aimed for a slightly overdamped response so the system would not oscillate into the dead zone. For NMPC we used a small offset in the gap reference so the motor stays a bit ahead of the load which helps keep the shaft engaged.

Challenges

The backlash nonlinearity was the biggest difficulty. The controller often sees zero torque inside the gap which makes the model and the real plant not match well. Simulation also needed smaller time steps because the switch between engaged and disengaged happens fast. Tuning NMPC was also tricky because changing one weight changed the whole behavior, requiring trade-off between aggressively minimizing the tracking error and staying within the torque limits and rate limits.

What We Learned

We discovered that model fidelity is essential for high performance control in mechanical systems with clearance. Even small backlash substantially degraded closed-loop performance when the controller assumed continuous, linear stiffness. This was clearly demonstrated by the linear MPC's failure. We concluded that NMPC is superior for these systems since it can model piecewise dynamics.

Next Steps

We would test the controllers on hardware, try real time adaptive estimation of backlash size, look for faster NMPC methods, and maybe extend the work to multi joint robot

systems. Investigating the use of feedforward torque or disturbance observers to further reduce the number of engagement and disengagement.

Contributions of team members

- Sneha Chandrasekaran: Nonlinear MPC design and simulation; 3D robot animation.
- Ashon Antony: Linear MPC design, tuning, and simulation.
- Brett Van Camp: LQR, second-order controller design; state observer design.
- DeAndre SanAgustin: System properties analysis (stability, controllability, observability) and controller comparison.
- All Members: Physical system modeling, mathematical model derivation, report writing, presentation preparation, and problem formulation.